CS 499 Milestone Three Narrative

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Computer Science Capstone

Artifact Description

The artifact selected for this milestone is a reusable Python CRUD module originally developed for the CS 340 Animal Shelter project. This module provides a structured interface for creating, reading, updating, and deleting records in a MongoDB collection. The artifact was previously enhanced for software design and engineering practices and now serves as the foundation for demonstrating algorithms and data structures concepts through controlled data handling and decision logic.

Justification for Inclusion and Enhancement

This artifact was selected for the enhancement of algorithms and data structures because it contains multiple decision paths and data transformations that benefit from clear, intentional algorithmic design. While the module does not implement complex mathematical algorithms, it relies heavily on structured control flow, input normalization, and consistent handling of dictionary-based data structures. These characteristics make it well suited for demonstrating practical algorithmic reasoning in a real-world backend context.

For this enhancement, the focus was placed on improving how input data and queries are processed before interacting with the database. Helper methods were introduced to normalize filter inputs and validate dictionary structures, ensuring that all CRUD operations operate on predictable and well-defined data. Centralizing this logic reduces duplicated conditional checks and creates a clearer algorithmic flow across methods. Additional refinements were made to simplify branching decisions in update and delete operations, improving readability and reducing unnecessary complexity while preserving existing functionality

Course Outcomes and Alignment

This enhancement aligns with the CS 499 course outcomes related to designing and evaluating computing solutions using appropriate algorithms and data structures. The updated module demonstrates the ability to analyze existing code, identify opportunities for improved control flow and data handling, and apply structured decision-making techniques to enhance correctness and maintainability. The enhancement follows the plan outlined in earlier milestones and remains consistent with the approved scope of the artifact.

Reflection on the Enhancement Process

Working on this enhancement reinforced the importance of algorithmic clarity even in application-level code. While the original implementation was functional, the enhancement process highlighted how small changes, such as normalizing inputs and reducing repeated branching logic, can significantly improve the readability and reliability of a system. One challenge was ensuring that improvements remained focused on algorithmic structure rather than introducing new features or altering behavior. This experience emphasized that effective algorithms are not limited to complex data structures, but are also present in how systems validate, transform, and process data consistently. These skills are essential for developing maintainable backend systems and will continue to inform my approach to future software development.